Component Technical Sheet

This document contains detailed technical information for each component used in the CubeSat prototype system.

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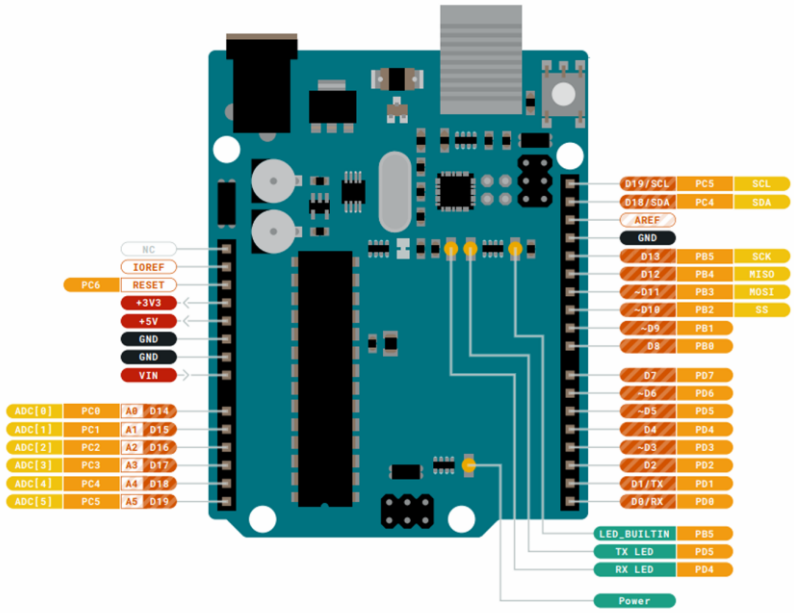
1. Arduino UNO R3



Pinout

Specifications

- ATmega328P MCU
- 5V operating voltage
- 16 MHz clock
- 32KB flash
- 20mA per I/O pin

Pinout

- Digital pins 0–13 (PWM on 3,5,6,9,10,11)
- Analog pins A0–A5
- UART, SPI, I2C
- Power: 5V, 3.3V, Vin, GND

Working Principle

Executes code via AVR microcontroller, handling ADC, timers, UART, SPI, I^2C interfaces.

Input/Output Signals

- Digital I/O: 0–5V
- Analog inputs: 0–5V mapped to 0–1023
- PWM outputs: 490–980 Hz

Operating Conditions

- Voltage: 7–12V recommended
- Temp: −40 to +85°C

Absolute Maximum Ratings

- I/O pin max: 5.5V
- Vcc max: 20V
- Total current max: 200mA

Electrical Characteristics

- Logic high: >3V
- Logic low: <1.5V
- ADC accuracy ±2 LSB

Typical Applications

Motor control, sensor interfacing, CubeSat avionics prototyping.

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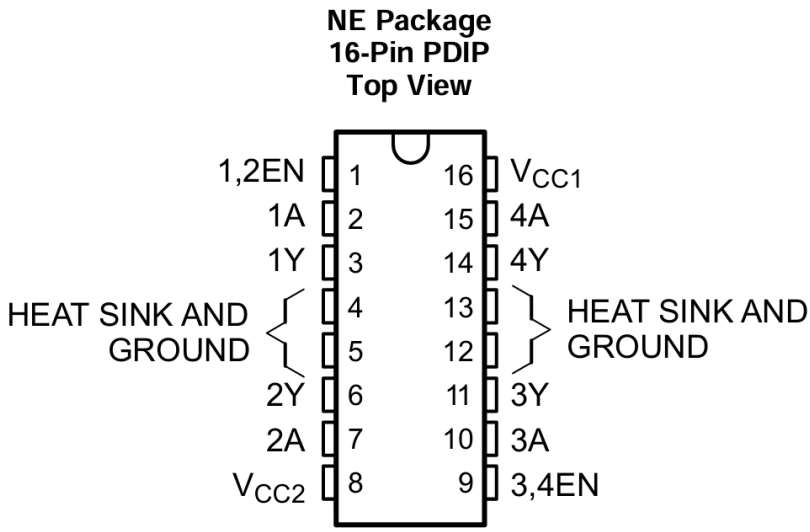
2. L293 Motor Driver



Specifications

- Dual H-bridge
- Motor voltage: up to 36V
- Logic: 5V
- 600mA/channel (1.2A peak)

Pinout

- EN1, EN2: enables
- Inputs: 1A,2A,3A,4A
- Outputs: 1Y–4Y
- Vs: Motor supply
- Vss: Logic supply

Working Principle

Controls DC motors by switching current direction using two H-bridges.

Input/Output Signals

- Logic input: 5V
- Output: Motor voltage level current up to 600mA

Operating Conditions

- Logic: 4.5–5.5V
- Motor: 4.5–36V

Absolute Maximum Ratings

- Output current peak 1.2A
- Temp: −40 to 150°C junction

Electrical Characteristics

- Input high >2.3V
- Voltage drop across outputs ~1.2–3V

Typical Applications

Motor control, robotics, CubeSat reaction wheel drivers.

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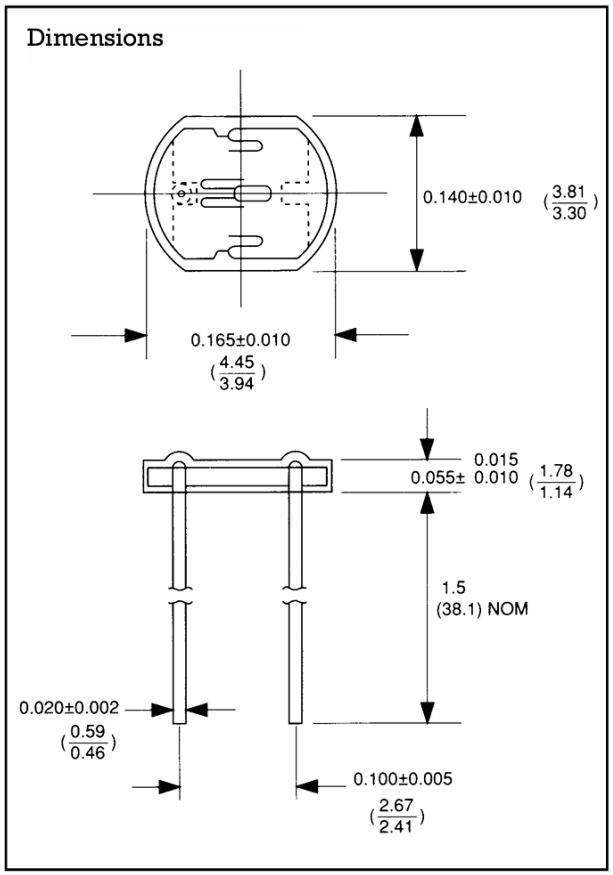
3. LDR (NORP12 / NSL19-M51)



Specifications
- 400Ω @ 1000 lux
- 9kΩ @ 10 lux
- 1MΩ dark resistance
- Rise: 2.8ms
- Fall: 48ms

Pinout
Two-terminal passive resistor, no polarity.

Working Principle
CdS photocell whose resistance decreases

with light intensity.

Input/Output Signals
- Input: Light intensity
- Output: Voltage (via divider)

Operating Conditions
- Max voltage 320V
- Temp: −60 to +75°C

Absolute Maximum Ratings
- 75mA max current

Electrical Characteristics
Logarithmic response curve, memory effect from storage lighting.

Typical Applications
Light detection, solar tracking, CubeSat sun-sensing.

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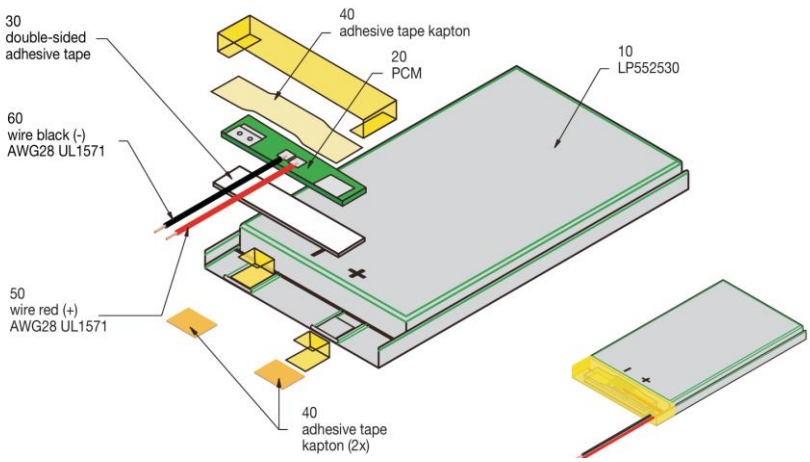
4. Li-Po Battery LP552530 400mAh



Specifications

- 400mAh
- 3.7V nominal
- 2.75–4.2V operating
- Charge current 200mA
- Discharge max 400mA

Pinout

Red = +, Black = –

Working Principle

Lithium-ion intercalation chemistry with PCM protection board.

Input/Output Signals

- Output: 3.0–4.2V depending on charge
- Input: CC/CV charging

Operating Conditions

- Charge: 0–45°C
- Discharge: −20–60°C

Absolute Maximum Ratings

- Overcharge cutoff: 4.275V
- Overdischarge cutoff: 2.75V

Electrical Characteristics

<200mΩ internal resistance

Typical Applications

Portable systems, CubeSat power modules.

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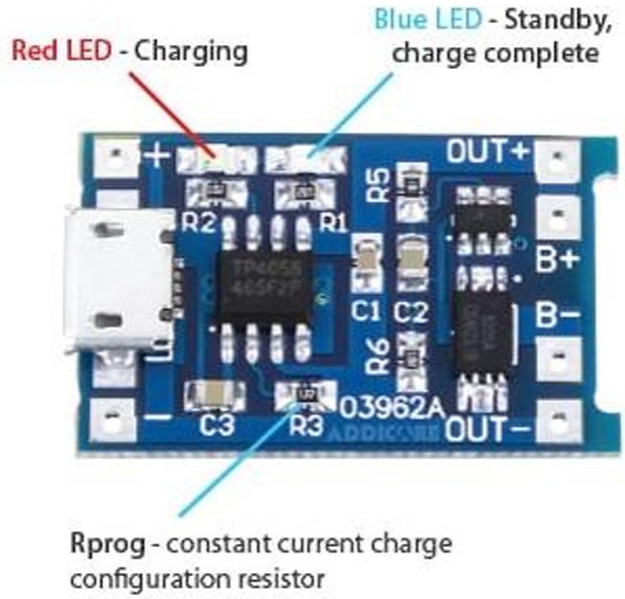

5. TP4056 Charger/Protection Module

Specifications

- 1A charge current
- 4.2V float
- Includes DW01A + FS8205A protection

Pinout

- IN+/IN−
- B+/B−
- OUT+/OUT−

Working Principle

CC/CV charging + integrated lithium protection circuitry.

Input/Output Signals

- RED LED = charging
- BLUE LED = done

Operating Conditions

Requires 5V input, 1A recommended.

Absolute Maximum Ratings

- Overcharge cutoff: 4.3V
- Overdischarge cutoff: 2.4V
- Overcurrent: 3A

Electrical Characteristics

- Trickle charge <2.9V at 130mA

Typical Applications

Battery charging in CubeSat power systems.

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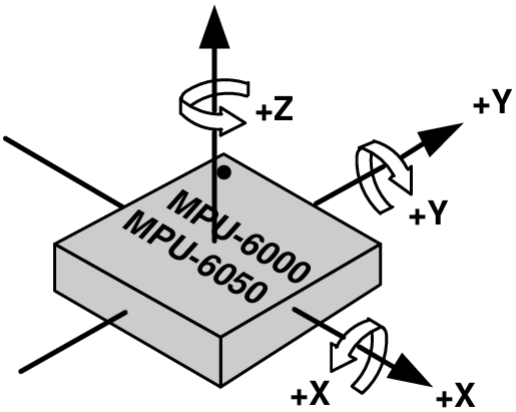
6. MPU6050 IMU



Specifications
- Gyro ranges ±250–2000 dps
- Accel ranges ±2–16g
- 16-bit ADCs

Pinout
- VDD
- VLOGIC
- SDA/SCL
- INT
- FSYNC
- REGOUT
- CPOUT

Working Principle
MEMS gyro + accelerometer with DMP for sensor fusion.

Input/Output Signals
- I^2C 400kHz
- INT signals data ready/motion
- Digital 16-bit sensor readings

Operating Conditions
- 2.375–3.46V supply
- −40 to +85°C

Absolute Maximum Ratings
- 6V max
- 10,000g shock

Electrical Characteristics
- Noise density: 400µg/√Hz accel

Typical Applications
Attitude sensing, reaction wheel stabilization in CubeSats.

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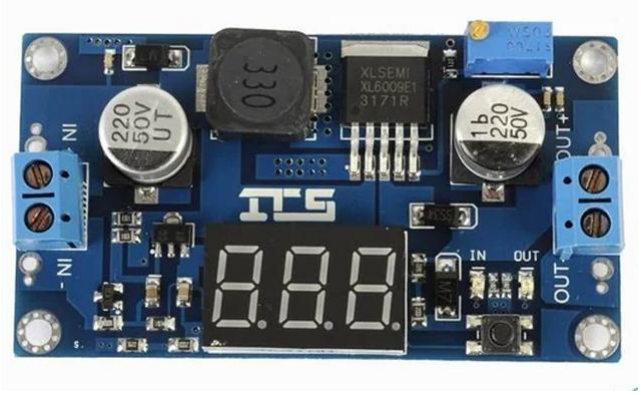
7. XL6009 Boost Converter Module



Specifications
- Input: 4.2–32V
- Output: 5–52V
- Up to 4A input
- 94% efficiency

Pinout
- IN+/IN−
- OUT+/OUT−
- Potentiometer for voltage
- Display button

Working Principle
Boost switching regulator increasing voltage using inductor + diode.

Input/Output Signals
- Input: DC voltage
- Output: higher DC regulated voltage

Operating Conditions
- −40 to +85°C

Absolute Maximum Ratings
- Dependent on input current limits (~4A)

Electrical Characteristics
- Ripple: ~50mV nominal

Typical Applications
Boosting 3.7V battery to 5V or higher for CubeSat payloads.